CLOCK & CALENDAR

Calendar

- 1. Finding the Day without Reference to a Given Day
- 2. Finding the Day with Reference to a Given day
- 3. Two Same Calendars
- 4. After or Before

Clock

- 1. Angle between Hour and Minute hand
- 2. Reflex Angle
- 3. Mirror and Water image
- 4. Finding Exact minutes at a Respective Angle
- Gaining & losing

POINTS TO BE REMEMBERED

1. In 12 hours, both hands coincide 11 times & both hands coincide 22 coincide 22 times in a day.
2. Both hands will be at 90 44 times in a day.
3. Similarly, both hands will be in opposite direction(180° apart) 22 direction(180° apart) 22 times in a day.
4. A year to be called a Leap year if divisible by 4 and for a Century, it for a Century, it should be divisible by 400.
5. Last day of a century must be Tuesday, Thursday, or Saturday.
6. Last day of a century cannot be Monday, Tuesday,

FINDING THE DAY WITHOUT REFERNCE = D + M + Y + A + B

Month Code (M)

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	4	4	0	2	5	0	3	6	1	4	6

Year Code (Y)

1900-1999	0
2000-2099	6
2100-2199	4
2200-2299	2

Day Code

0	Sat
1	Sun
2	Mon
3	Tue
4	Wed
5	Thu
6	Fri

PRACTICE

1. Mahatma Gandhi was born on 2nd Oct 1869. What was the day of the week?

2. What was the day of the week on 17th August 2010

3. What was the day of the week on 17th July 1776

FINDING THE DAY WITH REFERNCE

If 5th Jan 1991 was Saturday, what day of the week was it on 4th March 1992.

Sol = Number of days between 5th Jan 1991 and 4th March 1992 = (365–5) days of 1991 + 31 days of Jan + 29 days of Feb + 4 days of March = 424 days = 60 weeks and 4 days

4th March will be 4 days beyond Saturday

The required day will be Wednesday

PRACTICE

What day of the week was it on 5th November 1987, if it was Monday on 4th April 1988?

PRACTICE

What day of the week was it on 15th April 1996, if it was Monday on 15th Feb 1997?

Two Same Calendars

Divide the last two digits of the given year by 4 and find the remainder.

Remainder	Add
1	6
2	11
3	11
0	**28**

Note – After Adding the respective years will have the same Calendar for the Given year

PRACTICE

1. **2011**
2. **1972**
3. **2024**
4. **2001**
5. **1872**

After or Before

1. If Today is Saturday, what would be the day after 72 days?

Sol After dividing 72 by 7, the remainder is 2 days
Hence, 2 days after Saturday is Monday

2. If Today is Friday, what would be the day before 67 days?

Sol After dividing 67 by 7, the remainder is 4 days
Hence, 4 days before Friday will be Monday

PRACTICE

1. If today is Wed, find the day after 100 day

2. If today is Tue, find the day before 90 day

3. If today is Sun, find the day after 86 day

Angle b/w the hour & minute hand

Find the Angle when the clock shows 8.30?

Sol –

Step 1- Hour hand * 30 ° = 8 * 30 ° = 240°

Step 2 – Minute Hand / 2 = 30 / 2 = 15 °

Step 3 – Minute Hand * 6 ° = 30 * 6 = 180 °

Angle = |Step 1+ Step 2 – Step 3|

Required Angle = | 240 °+ 15 °| – 180 ° = 75 °

PRACTICE

1. 6.20
2. 3.30
3. 11.25
4. 4.20
5. 12.40

Reflex Angle b/w the hour & minute hand

Find the Angle when the clock shows 8.30?

Sol -

Step 1- Hour hand * 30 ° = 8 * 30 ° = 240°

Step 2 - Minute Hand / 2 = 30 / 2 = 15 °

Step 3 - Minute Hand * 6 ° = 30 * 6 = 180 °

Angle = |Step 1+ Step 2 - Step 3|

Required Angle = | 240 °+ 15 °| - 180 ° = 75 °

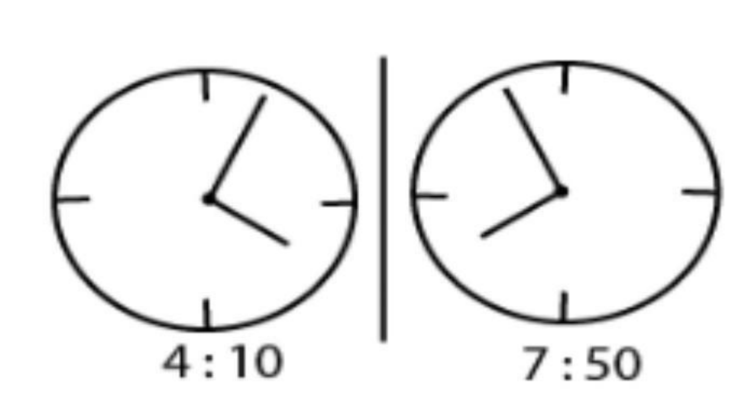
Reflex angle = 360 - 75 = 285°

PRACTICE

1. **6.20**
2. **3.30**
3. **11.25**
4. **4.20**
5. **12.40**

Mirror Image

Find the Mirror image when the clock shows 4.10.



For each clock-based mirror image , take the specified time and subtract it from 11:60 hours.

Water Image

Find the Water image when the clock shows 4.

For each clock-based water image , take the specified time and subtract it from 17:90 hours.

PRACTICE

1. Find the Mirror Image of 3.20, 12.45 and 6.30

2. Find the Water Image of 7.20, 12.55 and 9.30

Finding Exact minutes to a Respective Angle

At what time between 9 to 10 will the hands of a watch be together?

Sol-

Step 1- Multiply Hour hand with 30

Step 2- Add the required angle

Step 3- Multiply the sum by 2.

Step 4- Divide the value by 11 and find Quotient and Remainder.

According to the process, we get ((9*30)+ 0)*2= 540/11(Quotient – 49 and remainder is 1.

The final answer is 9.49 (1/11)

PRACTICE

1. At what time between 4 to 5 will the hands of a watch be opposite to each other

2. At what time between 7 to 8 will the hands of a watch be at 90 degree?

3. At what time between 2 to 3 will the hands of a watch be at 60 degree?

Gaining & Losing

1. A clock gains 15 minutes per day. It is set right at 12 noon. What time will it show at 4 am in the next day

Gaining & Losing

2. A clock loses 2 seconds every minute. If the clock was set to the correct time at 10 in the morning, find the time shown by the clock when the correct time is 5pm

Thank You
By
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